

Fraud Detection System — Documentation

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Overview

This project simulates a basic transaction monitoring system designed to detect suspicious financial activities. The system tracks users performing transactions from various locations and generates alerts based on predefined rules to flag potentially risky behavior. This setup can be useful for fraud detection or compliance monitoring.

Database Structure

Users Table

Stores basic user details.

Column	Type	Description
user_id	SERIAL PRIMARY KEY	Unique user ID
name	VARCHAR(100)	User's full name
email	VARCHAR(100)	User's email address
registered_at	TIMESTAMP	Timestamp of registration

Locations Table

Holds information about transaction locations.

Column	Type	Description
location_id	SERIAL PRIMARY KEY	Unique location ID
city	VARCHAR(100)	City name
ip_address	VARCHAR(50)	IP address for the location
country	VARCHAR(100)	Country name

Transactions Table

Tracks all user transactions.

Column	Type	Description
txn_id	SERIAL PRIMARY KEY	Unique transaction ID
user_id	INT	References user_id
amount	DECIMAL(12,2)	Transaction amount
txn_type	VARCHAR(20)	Transaction type (e.g., withdrawal, deposit)
txn_time	TIMESTAMP	Timestamp of transaction
location_id	INT	References location_id

Alerts Table

Records flagged transactions based on suspicious activity rules.

Column	Type	Description
alert_id	SERIAL PRIMARY KEY	Unique alert ID
txn_id	INT	References the flagged transaction
alert_type	VARCHAR(100)	Description of the alert
created_at	TIMESTAMP	Timestamp when alert was created

Suspicious Activity Detection Rules

- **High-value transactions:** Flag any transaction over \$10,000.
- **Multiple transactions within the same minute:** Flag users who perform more than 2 transactions in any single minute.
- **Transactions from multiple countries:** Flag users who have transactions from 3 or more different countries.
- **Withdrawal bursts:** Flag users with more than 3 withdrawal transactions within any rolling 10-minute window.

Sample Data

- 3 users: Alice, Bob, and Charlie.
- 5 locations across different cities and countries.
- Various transactions simulating normal and suspicious patterns.

Sample Alert Output

Alert ID	User	Amount	Transaction Time	Country	Alert Type	Alert Created
1	Alice	15000.00	2025-07-06 10:03:00	USA	High-value transaction	2025-07-06 10:03:00
2	Alice	400.00	2025-07-06 10:02:00	USA	Multiple transactions in the same minute	
3	Alice	300.00	2025-07-06 10:00:00	USA	Multiple transactions in the same minute	
4	Alice	250.00	2025-07-06 10:01:00	USA	Multiple transactions in the same minute	
5	Alice	500.00	2025-07-06 11:00:00	UK	Multiple countries detected	2025-07-06 11:00:00
6	Alice	200.00	2025-07-06 11:10:00	Japan	Multiple countries detected	
7	Alice	100.00	2025-07-06 11:20:00	India	Multiple countries detected	
8	Bob	95.00	2025-07-06 12:01:30	Germany	Withdrawal burst detected (10 min window)	2025-07-06 12:01:30
9	Bob	90.00	2025-07-06 12:00:00	Germany	Withdrawal burst detected (10 min window)	
10	Bob	80.00	2025-07-06 12:00:30	Germany	Withdrawal burst detected (10 min window)	
11	Bob	70.00	2025-07-06 12:01:00	Germany	Withdrawal burst detected (10 min window)	

Table 1: Sample alerts generated by the system

Summary

This project demonstrates how to monitor transaction data in real time and raise alerts on suspicious behavior using simple SQL queries and rules. The schema is modular and can be expanded for more complex fraud detection or compliance requirements. It's a solid foundation for building a financial transaction monitoring solution.